CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

PROJECT CHARTER & WORK PLAN

Student / Group Template

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
Group Members	Freek Schaefer-Gerber, Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Final Project Management
Submission Date	29 October 2026

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Project Overview

Provide title, background and business/research need.

Small IT support teams often manage customer requests through shared inboxes, spreadsheets, or informal messaging. This leads to lost tickets, unclear ownership, and a lack of visibility into backlog and priority. SmartDesk was developed to address this need by providing a centralised, secure, and traceable web-based ticketing system tailored for a small team environment. The system ensures customer data isolation and streamlines support workflows.

Aim and Objectives

Record the approved aim and objectives.

- Aim: To design, develop, and evaluate a web-based customer support ticketing prototype that centralises customer requests and improves workflow visibility.
- Objectives:

1. Document functional and non-functional requirements for three roles (Customer, Agent, Admin).
2. Design a normalised relational database and system architecture.
3. Develop a working application (using Python/Flask and SQLite) supporting authentication, ticket creation, assignment, and comments.
4. Implement strict access controls, validation, and a summary dashboard.
5. Test the prototype against documented requirements using automated and manual test cases.

Scope and Deliverables

List project boundaries and major outputs.

- In Scope: User registration/login (hashed passwords), role-based access control, customer data isolation, ticket creation and validation, staff-only assignment, status/priority management, attributable comments, status dashboard, and security measures (CSRF, SQL injection prevention).
- Out of Scope: Email notifications, file attachments, SLA timers, audit history tables, and enforced status transitions.
- Deliverables: Working Flask web application, SQLite database schema, SRS, complete dissertation, Git repository with commit history, and live presentation/demo.

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Team and Roles

Assign project roles and responsibilities.

- Freek Schaefer-Gerber (Project Lead / Database & Integration): Planning, primary supervisor contact, database design (ERD, SQLite), core Flask integration, and authoring Chapters 1-3.
- Wian Sandenbergh (Development & Testing Lead): Git maintenance, Flask templates, input validation, running pytest suites, defect logging, and authoring Chapters 4-6.
- Shared: Requirements gathering, system design review, documentation proofreading, and demo rehearsal.

Milestones

Identify major weekly milestones and expected completion dates.

- Week 1 (Proposal & Planning): Topic approved, SRS drafted, initial architecture designed.
- Week 2 (Development): Flask environment set up, database schema created, core ticket and auth routes built.
- Week 3 (Testing & Documentation): 3 automated and 20 manual tests executed, 5 defects fixed, chapters drafted.
- Week 4 (Finalisation): Dissertation merged and formatted, presentation rehearsed, final submission (29 Oct 2026).

Risks and Assumptions

Summarise major risks, assumptions and constraints.

- Risk: Outdated SRS describing the wrong tech stack (C#/SQL Server) causing evaluation confusion. *Mitigation:* SRS and Chapters were updated to reflect the final Python/Flask/SQLite implementation.
- Risk: Corrupted database during testing. *Mitigation:* Re-ran `flask init-db` and `seed-demo` scripts.
- Assumption: Local deployment with synthetic data is sufficient for the MVP demonstration. No real user data will be collected.

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Approval

Supervisor and group representative sign-off.

- Group Representative: *Freek Schaefer-Gerber* Date: 22-10-2026
- Supervisor: ____  _____ Date: 23-10-2026

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